

# An Autoencoder-Based Hybrid Detection Model for Intrusion Detection With Small-Sample Problem

Nan Wei, Lihua Yin<sup>✉</sup>, Jingyi Tan, Chuhong Ruan, Chuang Yin, Zhe Sun<sup>✉</sup>, and Xi Luo<sup>✉</sup>

**Abstract**—Cyber-attacks have become more frequent, targeted, and complex as the exponential growth in computer networks and the development of Internet of Things (IoT). Network intrusion detection system (NIDS) is an important and essential tool to protect network environments. However, the low performance of a NIDS against small malicious samples has seriously threatened the security of networks, thus directly leading to the loss of personal property and national interests. Given this, we propose an auto encoder-based hybrid detection model, abbreviated as AHDM, for the intrusion detection with small-sample problem. AHDM has a dual classifier framework. It trains first neural network based on the encoding features obtained from the autoencoder feature enhancement algorithm to detect small-sample malicious traffic. It trains second neural network using the original features to detect normal traffic and large-sample malicious traffic. The final detection result of malicious traffic is obtained by combining the detection results of the two neural networks. In experiments, we use three classic datasets (KDD CUP 99, CIC-IDS-2017, and IOT-23) and simulate the malicious traffic detection targeting extremely small-sample malicious traffic. The results show that AHDM has a higher detection rate for small-sample malicious traffic compared to the advanced detection models (DNN and ACID). In the IOT-23 dataset, the AHDM model shows an absolute advantage in detecting DDoS type of malicious traffic, with a detection rate of 0.71, which is much higher than the DNN (0.14) and ACID (0.14) models.

**Index Terms**—Malicious traffic detection, autoencoder, IOT-23, network intrusion detection system, small-sample dataset.

## I. INTRODUCTION

THE INCREASING adoption of Internet of Things (IoT) technology presents businesses with new opportunities to generate value and enhance integration in industries such as transportation, energy, and power systems [1]. However, with

Manuscript received 20 May 2023; revised 2 September 2023; accepted 10 November 2023. Date of publication 17 November 2023; date of current version 15 April 2024. This work was supported in part by the National Key R&D Program of China (No. 2022YFB3104100), the Guangdong Basic and Applied Basic Research Foundation (No. 2021A1515012454), the Project of Guangzhou Association for Science and Technology (No. 202201010609), the Young Talent Support Project of Guangzhou Association for Science and Technology (No. QT20220101122), and the Major Research Plan of the National Natural Science Foundation of China (No. 92167203). The associate editor coordinating the review of this article and approving it for publication was M. Conti. (Corresponding author: Lihua Yin.)

Nan Wei, Lihua Yin, Chuhong Ruan, Chuang Yin, Zhe Sun, and Xi Luo are with the Cyberspace Institute of Advanced Technology, Guangzhou University, Guangzhou 510006, Guangdong, China (e-mail: joey\_weinan@126.com; yinlh@gzhu.edu.cn; RUAN1597438160@163.com; yinnchuang@163.com; sunzhe@gzhu.edu.cn; luoxi0930@gzhu.edu.cn).

Jingyi Tan is with the Department of Information Resource Management, Guangzhou Xinhua University, Guangzhou 510520, Guangdong, China (e-mail: jytan1204@163.com).

Digital Object Identifier 10.1109/TNSM.2023.3334028

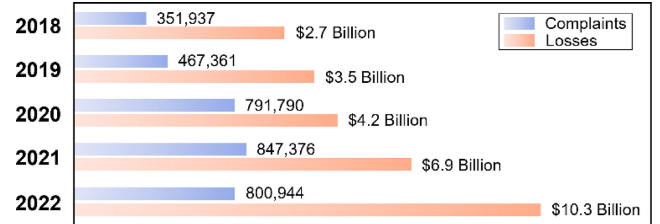


Fig. 1. Complaints and Losses from 2018 to 2022.

the continuous expansion of IoT devices and technology, there is a heightened risk of compromising sensitive information due to constant connectivity and data sharing [2]. These risks are further compounded by the development of 5G technology, which enables faster and more efficient communication between devices but also increases the potential for cyberattacks and security breaches [3]. Therefore, it is crucial for enterprises to prioritize cybersecurity measures and implement robust strategies to safeguard their data and networks against potential threats. Fig. 1 shows the Complaints and Losses over last five years, statistics from the Federal Bureau of Investigation's annual Internet Crime Complaint Center (IC3) report. In 2022, cybercrime costed Americans \$10.3 billion, a significant increase from \$6.9 billion in 2021, indicating a 49% increase [4]. This highlights the urgent need for businesses to take strong steps to protect themselves from cyberattacks and financial losses. Given this, various studies have been actively conducted to prevent potential network threats.

Network intrusion detection systems (NIDSs) provide protection through real-time monitoring and analysis of the system communication behavior [5]. Traditional NIDSs applied the misuse detection to identify network attacks. The misuse detection compared the signature (e.g., a sequence of bytes in the payload) of the network traffic with a pre-established threat dataset [6]. Such detection methods are then proved to be ineffective in the complex environment of IoT system due to it is computationally inefficient to collect the signature of the target traffic and compare it with a large-scale threat dataset, especially for the resource-constrained IoT devices [7].

An alternative strategy is malicious traffic detection, which monitors and analyzes the features of network traffic to identify normal or malicious traffic [8]. This strategy can dominate the misuse detection as it can detect unknown attacks that have distinct behaviors from the normal ones. Particularly, recent advance in machine learning (ML) facilitates the development

of malicious traffic detection techniques, as state-of-the-art ML models can distinguish normal and abnormal network activities with higher accuracy and less restriction [9]. This is mainly because the ML-based techniques have a strong capability of extracting insights or useful knowledge from the data in a timely and intelligent way, which allows themselves discover hidden patterns in large amount of network traffic boosting intrusion detection accuracy. However, these data driven-based NIDSs require large samples. Constrained by the high acquisition and labeling cost of traffic data, it is difficult to integrate a large number of labeled traffic data from real world network systems [10]. The training and updating of ML-based NIDSs can only use small-sample malicious traffic data, e.g., the amount of malicious traffic data is less than 100 or even less, which makes the ML-based NIDSs receiving low detect rates. Thus, it is essential to design a ML-based malicious traffic detection model using small-sample malicious traffic to detect the network intrusion.

Recent studies have proposed a range of few-shot learning methods for detecting malicious traffics with small-sample dataset [11]. These methods train the ML-based detection models with sufficient dataset from source network and fine tune the models with few shots from target network. However, it is often difficult to realize such an assumption in IoT networks where threat type always varies with the nonstationary environment. In the situation, NIDS with small-sample data can be described as a ML-based model which is treated with the sensors network data and may be not applicable to the network of actuators or appliances [12]. Furthermore, the attack distribution usually varies with network scale, protocol, and attack vector. The low performance of a NIDS against small-sample data has seriously threatened the security of IoT systems, thus directly leading to the loss of personal property and national interests. Therefore, the NIDS needs to be trained with the small-sample malicious traffic data of single network and ensure high precision.

To address the above-mentioned problem, we propose an autoencoder-based hybrid detection model for the intrusion detection. Additionally, to validate the detection performance of the proposed model, we design a craft case study with three classic datasets (KDD CUP 99, CIC-IDS-2017, and IOT-23), and compare the proposed system with advanced NIDSs in detection accuracy and generalization ability. In a nutshell, the contributions of this paper can be summarized as follows:

I. We develop an autoencoder-based feature enhancement (AFE) algorithm that use autoencoders to enhance the feature description of small-sample malicious traffic, thereby improving the detection accuracy of malicious traffic. This method groups normal and malicious traffic features based on Gaussian features, and uses autoencoders to encode the features in each group, and the obtained encoding features are used as input data for the model to detect small-sample malicious traffic.

II. We propose an autoencoder-based hybrid detection model, abbreviated as AHDM, for network intrusion detection. This model trains first neural network based on the encoding features obtained from the AFE algorithm to detect small-sample malicious traffic. It trains second neural network using the original features to detect normal traffic and large-sample

malicious traffic. The final detection result of malicious traffic is obtained by combining the detection results of the two neural networks.

III. A case study with three classic datasets (KDD CUP 99, CIC-IDS-2017, and IOT-23) are designed delicately for validating the performance of AHDM. Considering the effect of small-sample malicious traffics, we simulate the intrusion detection with small-sample dataset. Moreover, the proposed solution is compared with advanced NIDSs, such as deep learning neural networks (DNN) and adaptive clustering-based intrusion detection (ACID), for revealing its detection accuracy and generalization ability.

## II. RELATED WORK

This section briefly reviews the recent published papers relevant to the ML-based intrusion detection models for IoT networks and the malicious traffic detection methods targeting small-sample dataset, and highlights their shortcomings.

### A. ML-Based Intrusion Detection Models for IoT Networks

Various ML-based models have been proposed to propound intrusion detection mechanisms to efficiently detect and identify intruders in IoT systems [13]. Shukla [14] presented three intrusion detection methods based on K-means clustering, decision tree, and an ensemble of these two classical ML algorithms. The proposed approach is lightweight and is capable of accurately detecting wormhole attacks targeting IoT. Anthi et al. [15] focused on the intrusion detection of smart home IoT devices and proposed a three layers intrusion detection system (IDS) that uses a supervised approach to detect a range of popular network based cyber-attacks on IoT networks. In their work, several popular classifiers, such as naive Bayes, support vector machines, etc., were evaluated to detect seven network attack categories on a realistic testbed. Since DDoS attacks targeting IoT devices can cause severe corruption to the IoT metasystems, Jia et al. [16] proposed an LSTM-based IoT defense scheme termed FlowGuard for the detection of IoT DDoS attacks. The results indicated that the identification accuracy of the FlowGuard is as high as 98.9%. Similarly, Injadat et al. [17] and Moubayed et al. [18] proposed a decision tree (DT) based classification model and a genetic algorithm optimized random forest model to detect Botnet for protecting IoT networks against DDoS attacks.

Deep learning (DL) is considered an evolution of machine learning consisting of deeper neural networks. Ge et al. [19] and McDermott et al. [20] successively developed DL-based intrusion detection models, e.g., a feedforward neural network and a bidirectional long short term memory (LSTM) based recurrent neural network, for IoT networks. Meidan et al. [21] proposed a novel network-based anomaly detection method for the IoT called N-BaIoT that extracts behavior snapshots of the network and uses deep autoencoders (AE) to detect anomalous network traffic from compromised IoT devices.

It is clear that intrusion detection models based on machine learning techniques such as Bayesian networks and AE have become effective means of detecting IoT network attacks. However, these data-driven models require a large amount of

data to train the model. For IoT networks, it is difficult to obtain a large amount of training data. At the same time, the number of malicious traffic samples in IoT networks is much less than that of normal traffic, often leading to insufficient training of machine learning models and low detection rates.

### B. Malicious Traffic Detection Targeting Small-Sample Dataset

Many NIDSs have been designed by cybersecurity researchers to protect networks from attacks based on small-sample datasets. However, these existing NIDSs have failed to provide a high detection rate for small-sample datasets. To address this issue, Xiao and Xiao [22] proposed a simplified residual network consisting of several cascaded simplified residual blocks. This method has shown higher accuracy, recall, and F-score, especially for R2L and U2R attacks, compared to conventional residual networks. Zhang et al. [23] encoded network traffic data using P-Zigzag encoding scheme and detected cyber-attacks with a hybrid model based on convolutional neural network (CNN) and random forest to reduce the false alarm rate of malicious traffic detection with small-sample datasets. Maglaras et al. [24] built an intrusion detection system using data from a small-sample SCADA testbed, while Gao et al. [25] constructed an ensemble model that combined decision tree, random forest, and deep learning neural network for intrusion detection. Their experiment showed that optimizing the feature selection and preprocessing of intrusion detection data can greatly improve the detection rate.

Transfer learning technique can improve the accuracy of intrusion detection for the target domain that has fewer data by transferring the learned knowledge from a related source domain, which is also called few-shot network intrusion detection [26]. Chen et al. [27] trained a cross-domain DL-based model with imbalanced data and achieved high detection accuracy. Considering that malicious traffic data for IoT is expensive to collect and seldom available, several domain adaptation (DA) approaches were proposed to transfer the rich knowledge from the network intrusion domain to facilitate the intrusion detection for data-scarce IoT domains [28]. Since the network and IoT share several common attack types, by mapping both domains into a common feature subspace, these DA approaches can transfer the enriched network intrusion knowledge to assist intrusion detection in the target IoT domain [29]. Wu et al. [30] proposed a joint semantic transfer network toward effective intrusion detection for IoT network. By using a multisource heterogeneous DA algorithm, the proposed method obtained average a 10.3% of accuracy boost in their experiments.

However, the DA-based approaches rely heavily on the common attack types between source and target domains. For the IoT network, due to the many types of sensors and actuators, and the big differences in network traffic characteristics, these transfer models based on the DA algorithm are difficult to apply to the intrusion detection of the IoT network. Additionally, these published papers fail to validate the detecting performance of their proposed models targeting

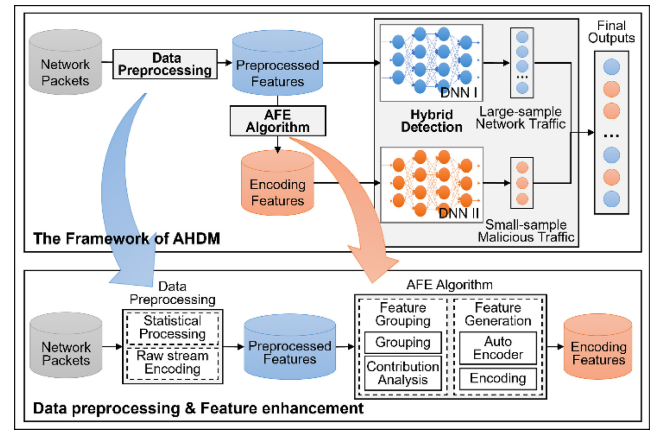


Fig. 2. The framework of AHDM.

extremely small-sample cyber-attacks, e.g., the extremely small-sample cyber-attacks (e.g., the amount of attack samples is less than 10), which always making intrusion detection more challenge.

## III. METHODOLOGY

We propose an autoencoder-based hybrid detection model, abbreviated as AHDM to detect the malicious traffic of network. This work aims at applying AHDM to detect the malicious traffic, particularly facing the situation of extremely small-sample cyber-attacks; and developing a lightweight detection model for quickly adapting to the resource constrained IoT devices. These aims are achieved by combining three key parts, including data preprocessing, AFE algorithm, and malicious traffic hybrid detection. AHDM's framework is shown in Fig. 2.

*I. Data preprocessing:* this step involves converting raw network packets into vectors of header and statistical features, and removing any invalid network features that could interfere with anomaly detection. Additionally, we will convert discrete variables into numeric values and normalize the continuous variables of raw features.

*II. AFE algorithm:* in this step, we analyze the statistical attributes of various traffic classes to identify subtle differences between normal and malicious traffic. We then encode features using autoencoder for detecting small-sample malicious traffic.

*III. Hybrid detection:* to improve detection accuracy and reduce false alarms, we construct a dual classifier that uses deep neural network (DNN).

The following sections will provide more details on each of these methods and show how their synergies lead to remarkable detection performance for malicious traffic.

### A. Data Preprocessing

AHDM processes large amounts of traffic by converting raw network packets into statistical features, which are then used for enhancing features and detecting malicious traffic. The data preprocessing involves two main steps. Firstly, statistical processing is done to remove duplicate streams from raw network packets and normalize continuous variables. Secondly, raw stream encoding is done to encode discrete

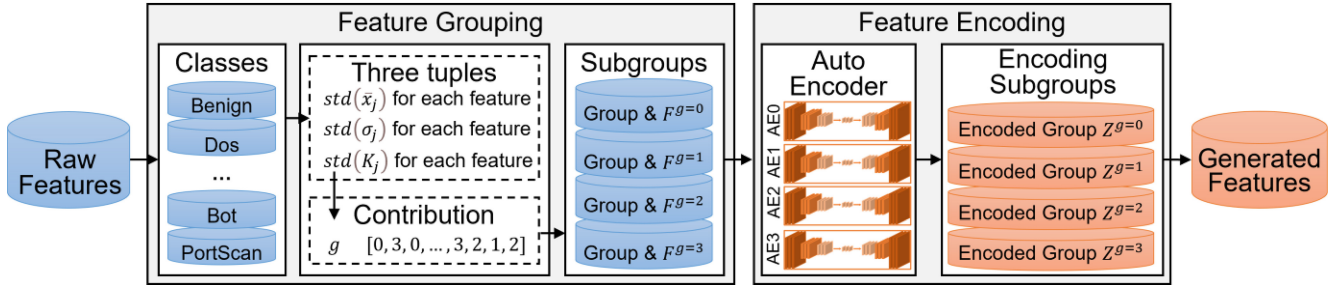


Fig. 3. Architecture of the AFE algorithm.

variables using one-hot encoding and convert labels to numeric values using the word2vec method [31]. These steps are the basic operations of the data preprocessing method, which may cause some latency depending on the capabilities of the device where AHDM is deployed.

### B. AFE Algorithm

We creatively propose an autoencoder-based feature enhancement method to describe the subtle difference between normal traffic and abnormal traffic. Compared with traditional clustering methods, this novel can screen out the clustering information from the features of traffic types, and describe the subtle difference between normal traffic and malicious traffic with low-dimensional embedded features. As shown in Fig. 3, this novel method implemented with two major steps: feature grouping and feature encoding.

*Step I:* the feature grouping is first proposed in our previous work [32], and here is applied to separate the raw features into four subgroups according to their statistical characteristics, providing grouped features for generating clustering features. Let  $x_{i,j}^l$  be value of  $i$ -th traffic sample and  $j$ -th feature.  $l = 1, 2, \dots, L$  denotes the traffic type, for instance,  $l = 1$  represents normal traffic,  $l = 2$  represents DOS traffic.  $i = 1, 2, \dots, M$  is the number of traffic samples,  $j = 1, 2, \dots, N$  is the number of traffic features. Let  $F = \{f_1, f_2, \dots, f_N\}$  be the features of traffic and  $f_j^l$  is the  $j$ -th feature of  $l$ -th type traffic. Then, we can calculate the three tuples of the features of each traffic type, including the coefficient of mean  $\bar{x}_{j,j}^l$ , standard deviation  $\sigma_j^l$ , and skewness  $S_j^l$ , using the follow equations:

$$\bar{x}_{j,j}^l = \frac{1}{m} \sum_i x_{i,j}^l \quad (1)$$

$$\sigma_j^l = \sqrt{\frac{1}{m-1} \sum_{i=1}^m (x_{i,j}^l - \bar{x}_{j,j}^l)^2} \quad (2)$$

$$S_j^l = \frac{1}{m} \sum_{i=1}^m \frac{(x_{i,j}^l - \bar{x}_{j,j}^l)^3}{(\sigma_j^l)^3} \quad (3)$$

After that, we use the standard deviation of the traffic type to illustrate the difference of  $j$ -th feature for traffic types, which can be describes as follows:

$$std_{j,t} = [std_{j,1}, std_{j,2}, std_{j,3}] \quad (4)$$

where  $t = \{1, 2, 3\}$  represents the coefficient of mean  $\bar{x}_{j,j}^l$ , standard deviation  $\sigma_j^l$ , and skewness  $S_j^l$ , respectively. The upper quartile  $q_{j,t} = [q_{j,1}, q_{j,2}, q_{j,3}]$  is used as a threshold of contribution index  $g_j$ .

$$g_{j,t} = \begin{cases} 1, & \text{where } std_{j,t} > q_{j,t} \\ 0, & \text{else} \end{cases} \quad (5)$$

where  $g_{j,t}$  is the contribution index of  $j$ -th feature and  $t$ -th tuple. The final contribution index of  $j$ -th feature can be obtained by  $g_j = 1 + \sum_{t=1}^T g_{j,t}$ , where the value of  $g_j$  ranges from 0 to 3. Consequently, the four subgroups of raw features ( $\tilde{x} = [\tilde{x}^0, \tilde{x}^1, \tilde{x}^2, \tilde{x}^3]$ ) can be obtained according to the contribution index of raw features. For instance,  $\tilde{x}^1$  involves the features that has a contribution index  $g_j = 1$ .

*Step II:* feature encoding is implemented by auto-encoder (AE). Suppose the input network features  $\tilde{x}^g$  belongs to  $n_g$  dimensional space and the encoding features  $z^g$  belongs to  $k_g$  dimensional space. Considering the contribution index of the features, we use four auto-encoders for each encoded feature group. The encoding dimensional  $k_g$  can be defined as:

$$k_g = 4 - g, \quad g = 0, 1, 2, 3 \quad (6)$$

An auto-encoder is a special and tricky three-layered neural network in which we set the output  $h_{W,b}(\tilde{x}^g) = (\hat{x}_1^g, \hat{x}_2^g, \dots, \hat{x}_m^g)^T$  equal to the input  $\tilde{x}^g = (\tilde{x}_1^g, \tilde{x}_2^g, \dots, \tilde{x}_m^g)^T$ .  $J$  is the reconstruction error. AE is an unsupervised learning algorithm and uses back propagation algorithm for training. The description of AE are as follows:

$$h_{W,b}(\tilde{x}^g) = g(f(\tilde{x}^g)) \approx \tilde{x}^g \quad (7)$$

$$J(W, b; \tilde{x}^g, z^g) = \frac{1}{2} \|h_{W,b}(\tilde{x}^g) - z^g\|^2 \quad (8)$$

where  $W$  and  $b$  are the weight and bias of the auto-encoder network. The final encoded features are combined as  $z^g = [z^0, z^1, z^2, z^3]$ .

### C. Malicious Traffic Hybrid Detection

The malicious traffic detection system utilizes a dual classifier that uses deep neural networks to process both raw and generated features. The output of the model determines the class to which the traffic belongs. The architecture of the malicious traffic detection model is illustrated in Fig. 4. The raw features are inputted into classifier I of the malicious traffic detection model to identify large-sample network traffic. The encoding features are inputted into classifier II of the

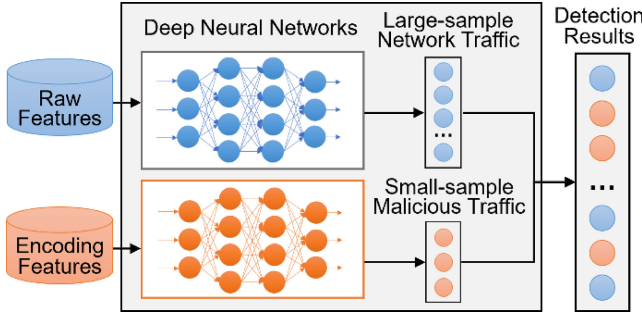


Fig. 4. Architecture of the hybrid detection model.

malicious traffic detection model to identify small-sample malicious traffic. Furthermore, the final detection results are obtained by combining the outputs of both detection models.

#### IV. IMPLEMENTATION

We ran AHDM on a 64-bit Windows 11 system with 32GB of RAM, 1TB of storage space, an 11th Gen Intel Core i7-11800H CPU running at 2.30GHz, and an NVIDIA GeForce RTX 3050Ti Laptop GPU with 16GB of memory. The detection model we used was implemented in Python 3.9 and relied on the numpy, pandas, sklearn, and PyTorch libraries. For the malicious traffic detection model, we used fully-connected neural networks with 3 hidden layers, each comprising 50, 30, and 10 neurons, respectively, for each classifier. The number of neurons in the input and output layers was equal to the number of features and the number of traffic types, respectively. We trained the neural networks using the Adam optimizer with a learning rate of 0.001.

#### V. PERFORMANCE EVALUATION

In this section, we first describe the datasets and evaluation metrics used in our performance evaluation. Then, we discuss how the feature enhancement method can magnify the subtle differences between normal and abnormal traffic, highlighting its superiority. Finally, we validate the performance of AHDM by comparing it with advanced NIDSs in the KDD CUP 99, CIC-IDS-2017, and IOT-23 datasets.

##### A. Evaluation Metrics

We utilize the testing data to calculate confusion matrices, which are then used to determine the values of True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) inferences. In addition, we employ five classic metrics to assess the quality of the classification results, including Accuracy, Precision, Recall, F-score (F1), and false alarm rate (FAR). As the number of traffic classes exceeds two and the distribution of raw traffic is imbalanced, we use weighted averaging for each metric.

$$Accuracy = (TP + FN) / (TP + FP + FN + TN) \quad (9)$$

$$Precision = TP / (TP + FP) \quad (10)$$

$$Recall = TP / (TP + FN) \quad (11)$$

$$F1 = 2 * Precision * Recall / (Precision + Recall) \quad (12)$$

$$FAR = FP / (FP + TN) \quad (13)$$

##### B. Datasets

Three well-known intrusion detection datasets, including KDD CUP 99, CIC-IDS-2017, and IOT-23, are applied to build network intrusion detectors [33], [34]. As the same as [31], the preprocessed dataset is then divided into a training and a testing set using a 7:3 split ratio. For validating the performance of proposed detection model targeting small-sample dataset, we synthesized seven training datasets with different percentage between normal and abnormal traffic samples for each dataset. The numbers of the dataset used in our experiment are shown in Table I.

##### C. Feature Enhancement

For visualizing the performance of feature enhancement method, we apply principal component analysis to extract the three principal components and plot the 3D scatter of the features (see Fig. 5).

The RF represents the raw features of network traffic instances. The CC means the clustering center of the feature clusters obtained by k-means algorithm, which is the same to the feature enhancement methods used in [31]. The CD applies the clustering distance between traffic sample and clusters, which is used in [35]. The AE represents the features obtained by auto-encoders. The AFE is the features refined by the AFE algorithm proposed in this work.

The visualized results reveal some interesting observations. For the dataset of KDD CUP 99, the differences between Normal and PROBE in RF are more obvious, but it is difficult to distinguish them from the space concentrated at (0,0,0) in DOS. The similarity between R2L and Probe sample spaces is high, making it difficult to distinguish them. After using AE, the distribution of network traffic features is similar to the original features, and there is no significant change in the differences between features. After using CC and CD features, the resolution of R2L samples with other network traffic features has been improved, but the overlap between other malicious traffic features and Normal features is high. The enhanced features obtained by using the AFE algorithm have significantly enhanced the differences between DOS and PROBE, and can clearly distinguish the spatial distribution of the two. R2L samples are distributed on the periphery of Normal samples, and R2L malicious traffic features can be easily identified by visualization.

The CIC-IDS-2017 dataset contains 10 types of malicious traffic features and Benign traffic features, totaling 11 categories. The Heartbleed sample features in the RF and AE features show a significant difference in distribution compared to other traffic features, making them easy to distinguish. The remaining traffic, including the Benign feature, has a more concentrated spatial distribution, making it difficult to distinguish between malicious and normal traffic. The use of CC and CD features has increased the spatial distribution difference between Benign and DOS samples, but the overall recognizability of the samples has not significantly improved. The use of AFE features has made the spatial distribution of network traffic samples more dispersed, with high recognition rates for DOS, BOT, DDoS, PortScan, Xss, and Benign samples, which is beneficial for improving the detection rate

TABLE I  
THE NUMBERS OF TRAFFIC CLASSES

| Dataset      | Classes       | Training data |       |       |       |       |       | Testing data |       |
|--------------|---------------|---------------|-------|-------|-------|-------|-------|--------------|-------|
|              |               | 1%            | 5%    | 10%   | 15%   | 20%   | 25%   | 30%          | -     |
| KDD CUP 99   | Normal        | 61744         | 61744 | 61744 | 61744 | 61744 | 61744 | 61744        | 26087 |
|              | DOS           | 589           | 3068  | 6476  | 10284 | 14569 | 19427 | 24976        | 16411 |
|              | Probe         | 24            | 122   | 256   | 405   | 575   | 766   | 987          | 625   |
|              | R2L           | 12            | 58    | 123   | 195   | 278   | 369   | 474          | 294   |
|              | U2R           | 2             | 4     | 6     | 13    | 17    | 22    | 28           | 12    |
| CIC-IDS-2017 | Benign        | 65341         | 65341 | 65341 | 65341 | 65341 | 65341 | 65341        | 27933 |
|              | Bot           | 2             | 14    | 30    | 49    | 69    | 92    | 119          | 438   |
|              | Brute Force   | 2             | 15    | 32    | 50    | 72    | 96    | 123          | 431   |
|              | DDoS          | 250           | 1307  | 2759  | 4383  | 6209  | 8279  | 10645        | 38536 |
|              | DOS           | 379           | 1979  | 4179  | 6637  | 9403  | 12538 | 16120        | 58192 |
|              | Heartbleed    | 1             | 1     | 1     | 1     | 1     | 1     | 1            | 4     |
|              | Infiltration  | 1             | 1     | 1     | 1     | 1     | 2     | 2            | 13    |
|              | Patator       | 18            | 94    | 199   | 316   | 449   | 598   | 769          | 2682  |
|              | PortScan      | 3             | 20    | 43    | 68    | 96    | 129   | 166          | 562   |
|              | Sql Injection | 1             | 1     | 1     | 1     | 1     | 1     | 1            | 6     |
|              | Xss           | 1             | 6     | 13    | 21    | 31    | 41    | 53           | 205   |
|              | Benign        | 1500          | 1500  | 1500  | 1500  | 1500  | 1500  | 1500         | 214   |
| IOT-23       | C&C           | 15            | 78    | 165   | 264   | 372   | 497   | 638          | 1228  |
|              | PortScan      | 1             | 2     | 2     | 2     | 3     | 3     | 4            | 7     |
|              | DDoS          | 1             | 1     | 1     | 2     | 2     | 2     | 2            | 3     |

of malicious traffic detection models. However, in the 3D distribution map, the Infiltration and Sql\_Injection types of malicious traffic samples have a small number, making it difficult to distinguish them visually.

Regarding to the dataset of IOT-23, IOT-23 dataset has a small sample size and an imbalance in the number of malicious traffic samples. The article mentions that the Benign, C&C, and PortScan malicious traffic samples are easier to distinguish using the RF, CC, CD, and AE features, but it is difficult to visually distinguish the small sample size of DDoS samples. However, in the AFE feature, DDoS is more recognizable.

Overall, using the AFE algorithm to obtain enhanced features compared to the original features, the spatial distribution recognizability of network traffic samples is more obvious, which is beneficial for enhancing the detection rate of malicious traffic detection models.

#### D. Network Intrusion Detection Results

According to Fig. 6, there are four models shown in three classic datasets, and their accuracy, precision, recall, F-score, and FAR are displayed in a radar chart. The models are deep learning neural networks (DNN), AFE algorithm combined with DNN (AFE+DNN), and adaptive clustering-based intrusion detection (ACID). In Fig. 6(a), for the KDD CUP 99 dataset, the ACID model performs poorly with an F-score of only 99.69% and a FAR of 0.07%. This indicates that using CC features for enhancement does not effectively improve the detection of malicious traffic. The DNN model, compared to ACID, shows an improvement of 0.02% to 0.03% in accuracy, precision, recall, and F-score while reducing the false alarm rate by 0.01%. Building upon this, incorporating the AFE algorithm into the DNN model results in a further 0.01% improvement in accuracy, precision, recall, and F-score but also increases the FAR by 0.01%. It is evident that while the AFE algorithm enhances features and

helps improve the detection of malicious traffic by increasing feature distinctiveness, it also leads to an increase in false alarms. The AHDM model achieves a prediction accuracy, precision, recall, and F-score of 99.76%, 99.75%, 99.76%, and 99.74%, respectively, all of which are higher than those of the DNN, AFE+DNN, and ACID models. Additionally, the AHDM model has a false alarm rate of only 0.06%, which is lower than that of the other three models. Therefore, by adopting a dual detection framework, it is possible to effectively combine the advantages of DNN and AFE+DNN, enhancing detection accuracy while reducing the model's FAR.

The same situation also occurred in the CIC-IDS-2017 dataset (Fig. 6 (b)). In this case, the AHDM model improved by 0.05%, 0.10%, 0.05%, and 0.07% respectively in terms of accuracy, precision, recall, and F-score compared to the DNN model; again, the false positive rate decreased by 0.01%. On the other hand, the ACID model decreased by 0.06%, 0.07%, 0.06%, and 0.07% respectively in terms of accuracy, precision, recall, and F-score compared to the DNN model; the false positive rate increased by 0.01%. Compared to the DNN model, the AFE+DNN model exhibits a slight increase in FAR, while precision has improved by 0.01%.

In the IoT-23 dataset, AHDM model shows significant improvements in accuracy, precision, recall, and F-score compared to the DNN model, with increases of 0.12%, 0.09%, 0.12%, and 0.17%, respectively; and the FAR has decreased by 0.04%. Although the overall performance of the ACID model is not as good as the AHDM model, it still shows a slight improvement compared to the DNN model, especially in terms of detection precision, with an increase of 0.09%. Overall, the detection performance of the AHDM model is better than that of the DNN, AFE+DNN, and ACID models, especially in the detection of malicious traffic in small-sample of IOT networks, with significant improvements in detection accuracy and false positive rates compared to traditional models. Although AFE algorithm can improve the accuracy, precision, recall, and

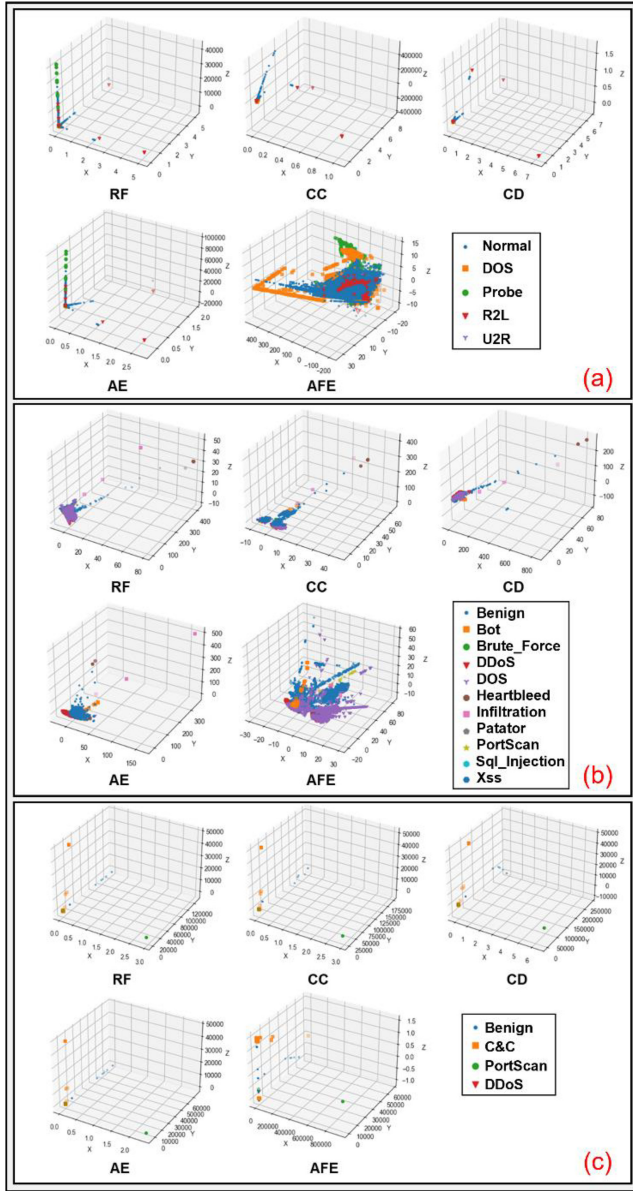


Fig. 5. Visualization of feature enhancement methods. (The datasets include: (a) KDD CUP 99, (b) CIC-IDS-2017, and (c) IOT-23).

F-score of DNN, but it may increase the FAR as well. Additionally, the ACID model is not as effective as the DNN model in KDD CUP 99 and CIC-IDS-2017, it performs better than the DNN model in detecting malicious traffic in small-sample of IOT networks.

Based on the sample ratio shown in Table I of the KDD CUP 99 dataset, it can be seen that the number of Normal and DOS samples is greater than 500, and the ACID, DNN, AFE+DNN, and AHDM models all have a detection rate over 99% for these two types of traffic. In Fig. 7, when malicious traffic accounts for 1%, the number of Probe, R2L, and U2R samples is less than 50, with 24, 12, and 2 respectively. For the U2R with the least number of samples, all four models used in this study were unable to successfully detect it. For R2L type of malicious traffic, the detection rate of the AHDM model was improved by 0.02 compared to DNN due to the use of the AFE algorithm. For PROBE, the detection rate of DNN,

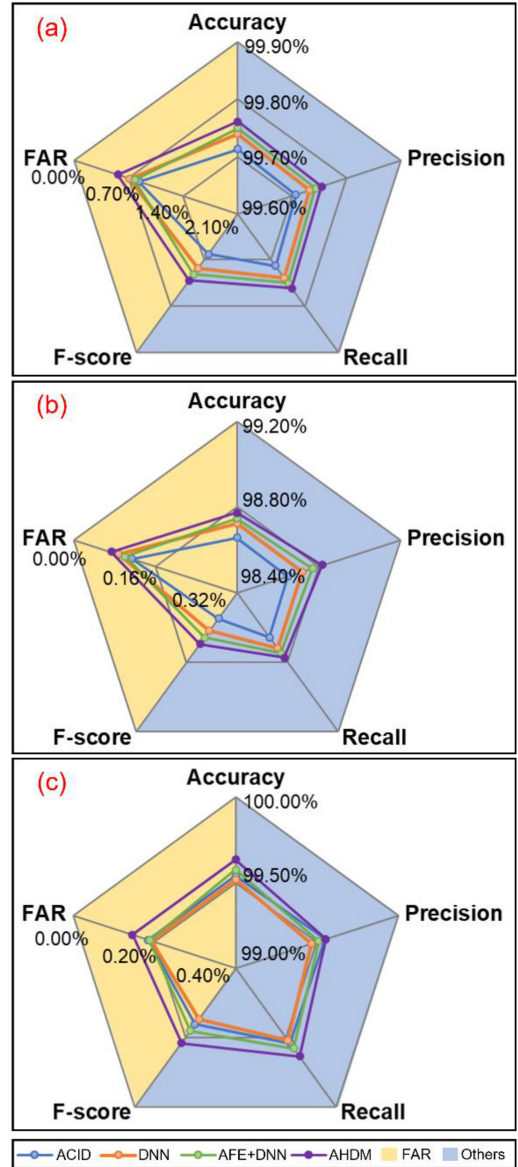


Fig. 6. Detecting performance of the models. (The datasets include: (a) KDD CUP 99, (b) CIC-IDS-2017, and (c) IOT-23).

ACID, and AHDM all reached 0.77. When malicious traffic accounts for 5%, the number of Probe, R2L, and U2R samples is 122, 58, and 4 respectively. The detection rates of AHDM for Probe, R2L, and U2R are 0.920, 0.840, and 0, respectively. In this case, when the number of malicious traffic samples is less than 5, the malicious traffic detection models are unable to successfully detect the malicious traffic. Additionally, by using AFE algorithm, 0.005 of the Probe type traffic are detected as DOS, which may increase the FAR of AFE+DNN model.

When malicious traffic accounts for 10%, the detection rates of ACID, DNN, AFE+DNN, and AHDM for U2R are 0.420. It proves that when the number of malicious traffic samples is greater than 5, the detection rate of the detection model can reach 0.420. When malicious traffic accounts for 15%, the number of U2R traffic samples is 13, and the detection rate of AFE+DNN and AHDM are 0.58, which are higher than that of DNN and ACID at 0.5. When malicious traffic accounts for

| Pct |        | ACID  |        |       |       |       | DNN   |        |       |       |       | AFE+DNN |        |       |       |       | AHDM  |        |       |       |       |
|-----|--------|-------|--------|-------|-------|-------|-------|--------|-------|-------|-------|---------|--------|-------|-------|-------|-------|--------|-------|-------|-------|
|     |        | DOS   | Normal | Probe | R2L   | U2R   | DOS   | Normal | Probe | R2L   | U2R   | DOS     | Normal | Probe | R2L   | U2R   | DOS   | Normal | Probe | R2L   | U2R   |
| 1%  | DOS    | 0.994 | 0.006  | 0     | 0     | 0     | 0.996 | 0.004  | 0     | 0     | 0     | 0.996   | 0.004  | 0     | 0     | 0     | 0.997 | 0.003  | 0     | 0     | 0     |
|     | Normal | 0     | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     | 0       | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     |
|     | Probe  | 0.042 | 0.18   | 0.778 | 0     | 0     | 0.04  | 0.19   | 0.77  | 0     | 0     | 0.042   | 0.19   | 0.768 | 0     | 0     | 0.04  | 0.19   | 0.77  | 0     | 0     |
|     | R2L    | 0     | 0.66   | 0     | 0.34  | 0     | 0     | 0.54   | 0     | 0.46  | 0     | 0       | 0.53   | 0     | 0.47  | 0     | 0     | 0.53   | 0     | 0.47  | 0     |
|     | U2R    | 0     | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     | 0       | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     |
| 5%  | DOS    | 0.998 | 0.002  | 0     | 0     | 0     | 0.998 | 0.002  | 0     | 0     | 0     | 0.999   | 0.001  | 0     | 0     | 0     | 0.999 | 0.001  | 0     | 0     | 0     |
|     | Normal | 0     | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     | 0       | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     |
|     | Probe  | 0.002 | 0.075  | 0.92  | 0     | 0     | 0.002 | 0.075  | 0.92  | 0     | 0     | 0.005   | 0.058  | 0.94  | 0     | 0     | 0.003 | 0.06   | 0.94  | 0     | 0     |
|     | R2L    | 0     | 0.2    | 0     | 0.8   | 0     | 0     | 0.18   | 0     | 0.82  | 0     | 0       | 0.16   | 0     | 0.84  | 0     | 0     | 0.16   | 0     | 0.84  | 0     |
|     | U2R    | 0     | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     | 0       | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     |
| 10% | DOS    | 1     | 0      | 0     | 0     | 0     | 0.999 | 0.001  | 0     | 0     | 0     | 1       | 0      | 0     | 0     | 0     | 1     | 0      | 0     | 0     | 0     |
|     | Normal | 0     | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     | 0       | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     |
|     | Probe  | 0     | 0.038  | 0.962 | 0     | 0     | 0     | 0.04   | 0.96  | 0     | 0     | 0       | 0.038  | 0.962 | 0     | 0     | 0     | 0.037  | 0.963 | 0     | 0     |
|     | R2L    | 0     | 0.14   | 0     | 0.86  | 0.003 | 0     | 0.12   | 0     | 0.877 | 0.003 | 0       | 0.085  | 0     | 0.91  | 0.003 | 0     | 0.095  | 0     | 0.9   | 0.003 |
|     | U2R    | 0     | 0.58   | 0     | 0     | 0.42  | 0     | 0.58   | 0     | 0     | 0.42  | 0       | 0.58   | 0     | 0     | 0.42  | 0     | 0.58   | 0     | 0     | 0.42  |
| 15% | DOS    | 1     | 0      | 0     | 0     | 0     | 1     | 0      | 0     | 0     | 0     | 1       | 0      | 0     | 0     | 0     | 1     | 0      | 0     | 0     | 0     |
|     | Normal | 0     | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     | 0       | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     |
|     | Probe  | 0.002 | 0.026  | 0.97  | 0     | 0     | 0.002 | 0.022  | 0.98  | 0     | 0     | 0.002   | 0.024  | 0.97  | 0     | 0     | 0.002 | 0.019  | 0.98  | 0     | 0     |
|     | R2L    | 0     | 0.11   | 0     | 0.89  | 0.003 | 0     | 0.099  | 0     | 0.9   | 0.003 | 0       | 0.1    | 0     | 0.89  | 0.003 | 0     | 0.068  | 0     | 0.93  | 0.003 |
|     | U2R    | 0     | 0.5    | 0     | 0     | 0.5   | 0     | 0.5    | 0     | 0     | 0.5   | 0       | 0.337  | 0     | 0.083 | 0.58  | 0     | 0.42   | 0     | 0     | 0.58  |
| 20% | DOS    | 1     | 0      | 0     | 0     | 0     | 1     | 0      | 0     | 0     | 0     | 1       | 0      | 0     | 0     | 0     | 1     | 0      | 0     | 0     | 0     |
|     | Normal | 0     | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     | 0       | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     |
|     | Probe  | 0     | 0.024  | 0.976 | 0     | 0     | 0     | 0.019  | 0.98  | 0     | 0     | 0       | 0.018  | 0.982 | 0     | 0     | 0     | 0.021  | 0.98  | 0     | 0     |
|     | R2L    | 0     | 0.058  | 0     | 0.939 | 0.003 | 0     | 0.051  | 0     | 0.946 | 0.003 | 0       | 0.054  | 0     | 0.946 | 0     | 0     | 0.054  | 0     | 0.946 | 0     |
|     | U2R    | 0     | 0.5    | 0     | 0     | 0.5   | 0     | 0.58   | 0     | 0     | 0.42  | 0       | 0.5    | 0     | 0     | 0.5   | 0     | 0.5    | 0     | 0     | 0.5   |
| 25% | DOS    | 1     | 0      | 0     | 0     | 0     | 1     | 0      | 0     | 0     | 0     | 1       | 0      | 0     | 0     | 0     | 1     | 0      | 0     | 0     | 0     |
|     | Normal | 0     | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     | 0       | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     |
|     | Probe  | 0     | 0.022  | 0.978 | 0     | 0     | 0     | 0.021  | 0.979 | 0     | 0     | 0       | 0.021  | 0.979 | 0     | 0     | 0     | 0.019  | 0.98  | 0     | 0     |
|     | R2L    | 0     | 0.065  | 0     | 0.93  | 0.003 | 0     | 0.061  | 0     | 0.94  | 0.003 | 0       | 0.034  | 0     | 0.96  | 0.003 | 0     | 0.017  | 0     | 0.98  | 0.003 |
|     | U2R    | 0     | 0.42   | 0     | 0     | 0.58  | 0     | 0.33   | 0     | 0     | 0.67  | 0       | 0.25   | 0     | 0.083 | 0.667 | 0     | 0.25   | 0     | 0.083 | 0.667 |
| 30% | DOS    | 1     | 0      | 0     | 0     | 0     | 1     | 0      | 0     | 0     | 0     | 1       | 0      | 0     | 0     | 0     | 1     | 0      | 0     | 0     | 0     |
|     | Normal | 0     | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     | 0       | 1      | 0     | 0     | 0     | 0     | 1      | 0     | 0     | 0     |
|     | Probe  | 0     | 0.014  | 0.986 | 0     | 0     | 0     | 0.014  | 0.986 | 0     | 0     | 0       | 0.014  | 0.986 | 0     | 0     | 0     | 0.014  | 0.986 | 0     | 0     |
|     | R2L    | 0     | 0.051  | 0     | 0.95  | 0     | 0     | 0.037  | 0     | 0.96  | 0     | 0       | 0.031  | 0     | 0.97  | 0     | 0     | 0.02   | 0     | 0.98  | 0     |
|     | U2R    | 0     | 0.083  | 0     | 0     | 0.917 | 0     | 0.083  | 0     | 0     | 0.917 | 0       | 0.083  | 0     | 0     | 0.917 | 0     | 0.083  | 0     | 0     | 0.917 |

Fig. 7. Normalized confusion matrixes of detection results in KDD CUP 99.

20%, AHDM's detection rate for R2L is 0.98, which is better than other three models. When malicious traffic accounts for 30%, the number of malicious traffic samples exceeds 25, and the detection effects of the three models are equivalent. The detection rates for malicious traffic are all higher than 0.9. Overall, although the AHDM model cannot detect malicious traffic when the number of malicious traffic samples is less than 5 in this dataset, its detection rate is higher than DNN and ACID when the number of malicious traffic samples is greater than 5. Again, by introducing dual classifier framework, the FAR of AHDM is lower than that of AFE+DNN model.

According to the flow sample ratio in Table I of the CIC-IDS-2017 dataset, there are 10 types of malicious traffic, which is more than the KDD CUP 99 dataset. In Fig. 8, when the malicious traffic accounts for 1%, the sample sizes of Bot, Brute\_Force, Heartbleed, Infiltration, PortScan, Sql Injection, and Xss are all less than 5. DNN, ACID, and AHDM can only detect Heartbleed attacks that have significant differences from other traffic features, but they are powerless against Bot, Infiltration, Sql Injection, and Xss malicious traffic. As for the Bot malicious traffic that DNN and ACID cannot recognize, the detection rate of AFE+DNN and AHDM model reach 0.1 and 0.3, respectively. At the same time, in terms of the detection rate of PortScan malicious traffic, AFE+DNN and AHDM are 0.07 higher than ACID. However, in this case, the false alarms caused by the adoption of the AFE algorithm are also quite evident. For instance, in the case of a larger sample size for DOS attacks, AFE+DNN exhibits a 0.02 decrease in detection rate compared to DNN, with 0.4% of them being incorrectly identified as DDoS attacks. Therefore, in this dataset, the AHDM model has an advantage over DNN,

| Models  |               | Benign | Bot | Brute_Force | DDoS  | DOS   | Heartbleed | Patator | PortScan | Xss   |
|---------|---------------|--------|-----|-------------|-------|-------|------------|---------|----------|-------|
| ACID    | Benign        | 1      | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Bot           | 0      | 1   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Brute_Force   | 0.956  | 0   | 0.044       | 0     | 0     | 0          | 0       | 0        | 0     |
|         | DDoS          | 0.007  | 0   | 0           | 0.993 | 0     | 0          | 0       | 0        | 0     |
|         | DOS           | 0.048  | 0   | 0           | 0     | 0.95  | 0          | 0       | 0        | 0     |
|         | Heartbleed    | 0      | 0   | 0           | 0     | 0     | 1          | 0       | 0        | 0     |
|         | Infiltration  | 1      | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Patator       | 0.21   | 0   | 0           | 0     | 0     | 0          | 0.79    | 0        | 0     |
|         | PortScan      | 0.63   | 0   | 0           | 0.027 | 0.041 | 0          | 0       | 0.3      | 0     |
|         | Sql_Injection | 1      | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
| DNN     | Xss           | 0.99   | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0.01  |
|         | Benign        | 1      | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Bot           | 0      | 1   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Brute_Force   | 0.956  | 0   | 0.044       | 0     | 0     | 0          | 0       | 0        | 0     |
|         | DDoS          | 0.006  | 0   | 0           | 0.99  | 0     | 0          | 0       | 0        | 0     |
|         | DOS           | 0.047  | 0   | 0           | 0     | 0.95  | 0          | 0       | 0        | 0     |
|         | Heartbleed    | 0      | 0   | 0           | 0     | 0     | 1          | 0       | 0        | 0     |
|         | Infiltration  | 1      | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Patator       | 0.077  | 0   | 0           | 0     | 0     | 0          | 0.92    | 0        | 0     |
|         | PortScan      | 0.56   | 0   | 0           | 0.027 | 0.039 | 0          | 0       | 0.37     | 0     |
| AFE+DNN | Sql_Injection | 1      | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Xss           | 0.98   | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0.019 |
|         | Benign        | 1      | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Bot           | 0.9    | 0.1 | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Brute_Force   | 0.95   | 0   | 0.044       | 0     | 0     | 0          | 0       | 0        | 0.005 |
|         | DDoS          | 0.006  | 0   | 0           | 0.99  | 0.005 | 0          | 0       | 0        | 0     |
|         | DOS           | 0.067  | 0   | 0           | 0.004 | 0.93  | 0          | 0       | 0        | 0     |
|         | Heartbleed    | 0      | 0   | 0           | 0     | 0     | 1          | 0       | 0        | 0     |
|         | Infiltration  | 1      | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Patator       | 0.059  | 0   | 0           | 0     | 0     | 0          | 0.94    | 0        | 0     |
| AHDM    | PortScan      | 0.56   | 0   | 0           | 0.027 | 0.043 | 0          | 0       | 0.37     | 0     |
|         | Sql_Injection | 1      | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Xss           | 0.98   | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0.019 |
|         | Benign        | 1      | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Bot           | 0.7    | 0.3 | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Brute_Force   | 0.956  | 0   | 0.044       | 0     | 0     | 0          | 0       | 0        | 0     |
|         | DDoS          | 0.006  | 0   | 0           | 0.99  | 0     | 0          | 0       | 0        | 0     |
|         | DOS           | 0.046  | 0   | 0           | 0     | 0.95  | 0          | 0       | 0        | 0     |
|         | Heartbleed    | 0      | 0   | 0           | 0     | 0     | 1          | 0       | 0        | 0     |
|         | Infiltration  | 1      | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Patator       | 0.04   | 0   | 0           | 0     | 0     | 0          | 0.96    | 0        | 0     |
|         | PortScan      | 0.56   | 0   | 0           | 0.027 | 0.043 | 0          | 0       | 0.37     | 0     |
|         | Sql_Injection | 1      | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0     |
|         | Xss           | 0.98   | 0   | 0           | 0     | 0     | 0          | 0       | 0        | 0.019 |

Fig. 8. Normalized confusion matrixes of detection results in CIC-IDS-2017.

AFE+DNN and ACID in detecting small-sample malicious traffic.

The IOT-23 dataset collects much less data than KDD CUP 99 and CIC-IDS-2017. In Fig. 9, the Benign has only 1500 samples, with C&C, PortScan, and DDoS having 15, 1,

| Pct |          | ACID   |     |      |          | DNN    |     |      |          | AFE+DNN |       |      |          | AHDM   |     |      |          |
|-----|----------|--------|-----|------|----------|--------|-----|------|----------|---------|-------|------|----------|--------|-----|------|----------|
|     |          | Benign | C&C | DDoS | PortScan | Benign | C&C | DDoS | PortScan | Benign  | C&C   | DDoS | PortScan | Benign | C&C | DDoS | PortScan |
| 1%  | Benign   | 1      | 0   | 0    | 0        | 1      | 0   | 0    | 0        | 1       | 0     | 0    | 0        | 1      | 0   | 0    | 0        |
|     | C&C      | 0      | 1   | 0    | 0        | 0      | 1   | 0    | 0        | 0       | 1     | 0    | 0        | 0      | 1   | 0    | 0        |
|     | DDoS     | 0.86   | 0   | 0.14 | 0        | 0.86   | 0   | 0.14 | 0        | 0.29    | 0     | 0.71 | 0        | 0.29   | 0   | 0.71 | 0        |
|     | PortScan | 0.67   | 0   | 0    | 0.33     | 0.67   | 0   | 0    | 0.33     | 0.33    | 0     | 0    | 0.67     | 0.33   | 0   | 0    | 0.67     |
| 5%  | Benign   | 1      | 0   | 0    | 0        | 1      | 0   | 0    | 0        | 1       | 0     | 0    | 0        | 1      | 0   | 0    | 0        |
|     | C&C      | 0      | 1   | 0    | 0        | 0      | 1   | 0    | 0        | 0       | 1     | 0    | 0        | 0      | 1   | 0    | 0        |
|     | DDoS     | 0.71   | 0   | 0.29 | 0        | 1      | 0   | 0    | 0        | 1       | 0     | 0    | 0        | 1      | 0   | 0    | 0        |
|     | PortScan | 0.33   | 0   | 0    | 0.67     | 0.33   | 0   | 0    | 0.67     | 0.33    | 0     | 0    | 0.67     | 0.33   | 0   | 0    | 0.67     |
| 10% | Benign   | 1      | 0   | 0    | 0        | 1      | 0   | 0    | 0        | 1       | 0     | 0    | 0        | 1      | 0   | 0    | 0        |
|     | C&C      | 0      | 1   | 0    | 0        | 0      | 1   | 0    | 0        | 0       | 1     | 0    | 0        | 0      | 1   | 0    | 0        |
|     | DDoS     | 0.71   | 0   | 0.29 | 0        | 0.71   | 0   | 0.29 | 0        | 0.29    | 0     | 0.71 | 0        | 0.29   | 0   | 0.71 | 0        |
|     | PortScan | 0.33   | 0   | 0    | 0.67     | 0.67   | 0   | 0    | 0.33     | 0.33    | 0     | 0    | 0.67     | 0.33   | 0   | 0    | 0.67     |
| 15% | Benign   | 1      | 0   | 0    | 0        | 1      | 0   | 0    | 0        | 1       | 0     | 0    | 0        | 1      | 0   | 0    | 0        |
|     | C&C      | 0      | 1   | 0    | 0        | 0      | 1   | 0    | 0        | 0.004   | 0.996 | 0    | 0        | 0      | 1   | 0    | 0        |
|     | DDoS     | 0.86   | 0   | 0.14 | 0        | 0.86   | 0   | 0.14 | 0        | 0.86    | 0     | 0.14 | 0        | 0.86   | 0   | 0.14 | 0        |
|     | PortScan | 0.33   | 0   | 0    | 0.67     | 0.33   | 0   | 0    | 0.67     | 0.33    | 0     | 0    | 0.67     | 0.33   | 0   | 0    | 0.67     |
| 20% | Benign   | 1      | 0   | 0    | 0        | 1      | 0   | 0    | 0        | 1       | 0     | 0    | 0        | 1      | 0   | 0    | 0        |
|     | C&C      | 0      | 1   | 0    | 0        | 0      | 1   | 0    | 0        | 0.004   | 0.996 | 0    | 0        | 0      | 1   | 0    | 0        |
|     | DDoS     | 0.71   | 0   | 0.29 | 0        | 0.71   | 0   | 0.29 | 0        | 0.29    | 0     | 0.71 | 0        | 0.29   | 0   | 0.71 | 0        |
|     | PortScan | 0.33   | 0   | 0    | 0.67     | 0.33   | 0   | 0    | 0.67     | 0.33    | 0     | 0    | 0.67     | 0.33   | 0   | 0    | 0.67     |
| 25% | Benign   | 1      | 0   | 0    | 0        | 1      | 0   | 0    | 0        | 1       | 0     | 0    | 0        | 1      | 0   | 0    | 0        |
|     | C&C      | 0      | 1   | 0    | 0        | 0      | 1   | 0    | 0        | 0       | 1     | 0    | 0        | 0      | 1   | 0    | 0        |
|     | DDoS     | 0.86   | 0   | 0.14 | 0        | 0.86   | 0   | 0.14 | 0        | 0.86    | 0     | 0.14 | 0        | 0.86   | 0   | 0.14 | 0        |
|     | PortScan | 0.33   | 0   | 0    | 0.67     | 0.33   | 0   | 0    | 0.67     | 0.33    | 0     | 0    | 0.67     | 0      | 0   | 0.33 | 0.67     |
| 30% | Benign   | 1      | 0   | 0    | 0        | 1      | 0   | 0    | 0        | 1       | 0     | 0    | 0        | 1      | 0   | 0    | 0        |
|     | C&C      | 0      | 1   | 0    | 0        | 0      | 1   | 0    | 0        | 0.004   | 0.996 | 0    | 0        | 0      | 1   | 0    | 0        |
|     | DDoS     | 0.71   | 0   | 0.29 | 0        | 0.71   | 0   | 0.29 | 0        | 0.29    | 0     | 0.71 | 0        | 0.29   | 0   | 0.71 | 0        |
|     | PortScan | 0.33   | 0   | 0    | 0.67     | 0.33   | 0   | 0    | 0.67     | 0.33    | 0     | 0    | 0.67     | 0.33   | 0   | 0    | 0.67     |

Fig. 9. Normalized confusion matrixes of detection results in IOT-23.

and 1 sample respectively when malicious traffic accounts for 1%. DNN, AFE+DNN, ACID, and AHDM can all achieve 100% detection rate for C&C type detection. This is because, as shown in Fig. 5(c), the distinguishability of C&C is high, even with a small number of samples, accurate detection can still be achieved. In terms of detecting DDoS type malicious traffic, AFE+DNN and AHDM model show absolute advantages, with the detection rate of 0.71, much higher than the DNN (0.14) and ACID (0.14) models.

When the malicious traffic accounts for 5%, the detection rate of ACID model for DDoS is 0.29, which is better than DNN and AHDM. When the malicious traffic accounts for 20% and 30%, the number of PortScan and DDoS samples are less than 5. The detection rate of DNN and ACID models for PortScan reaches 0.29, but is still lower than that of AFE+DNN and AHDM model (0.71). When the malicious traffic accounts for 25%, the detection rates of the four models for PortScan and DDoS are consistent, which are 0.14 and 0.67 respectively. However, at this time, the AHDM model will classify the remaining PortScan as DDoS in the detection results. Although the model cannot accurately identify the PortScan type, it can distinguish it from the Benign and determine it as malicious traffic. Furthermore, in the predictions produced by the AFE+DNN model, due to the influence of the AFE algorithm's enhanced features, when the proportion of malicious traffic is at 15%, 20%, and 30%, the detection rate for the C&C-type malicious traffic decreases by 0.4%. Building upon this, the dual-structured AHDM model effectively mitigates this risk.

Therefore, AHDM has a significant advantage over DNN, AFE+DNN, and ACID models in terms of detection rate for small-sample IoT network malicious traffic detection. Even when the number of malicious traffic samples is less than 5, it can achieve a high detection rate and effectively detect IoT network attack behaviors.

## VI. CONCLUSION AND FUTURE DIRECTIONS

This paper focus on the malicious traffics detecting with small-sample dataset. Given this, we propose an auto encoder-based hybrid detection model, namely AHDM, for the malicious traffics detection in IoT systems. This model trains first neural network based on the encoding features obtained from the autoencoder feature enhancement algorithm to detect small-sample malicious traffic. It trains second neural network using the original features to detect normal traffic and large-sample malicious traffic. The final detection result of malicious traffic is obtained by combining the detection results of the two neural networks. Additionally, for validating the performance of AHDM, we design three delicate case study with the classic datasets, including KDD CUP 99, CIC-IDS-2017, and IOT-23, and compare it with advanced network intrusion detection systems (NIDSs). The experimental results show that:

I. The detection performance of the AHDM model is superior to that of the DNN and ACID models, with significant improvements in detection accuracy, recall rate, and false alarm rate (FAR) compared to advanced models. In the IOT-23 dataset, the AHDM model shows significant improvements in accuracy, precision, recall rate, and F-score compared to the DNN model, with increases of 0.12%, 0.09%, 0.12%, and 0.17%, respectively. At the same time, the false alarm rate has decreased by 0.04%.

II. The AHDM model has a higher detection rate for small-sample malicious traffic compared to the DNN and ACID models. Based on the KDD CUP 99 dataset, when the malicious traffic accounts for 1% of the dataset, the R2L sample size is 12. For this type of malicious traffic, the AHDM model has detection rate improvements of 0.01 and 0.13 compared to the DNN and ACID models, respectively. In the CIC-IDS-2017 dataset, the AHDM model has a detection rate of 0.3 for the Bot that the DNN and ACID models

cannot detect. In the IOT-23 dataset, the AHDM model shows an absolute advantage in detecting DDoS type of malicious traffic, with a detection rate of 0.71, which is much higher than the DNN (0.14) and ACID (0.14) models.

However, we also found that in our case studies, when the number of malicious traffic sample is less than 5, the NIDSs we used are difficult to detect these small-sample malicious traffics. Thus, how to improve the detection rate of the NIDSs with such extremely small-sample malicious traffic is still an important issue for our future research.

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**Nan Wei** received the Ph.D. degree from Southwest Petroleum University in 2020. He is currently a Lecturer of Network Security with the Cyberspace Institute of Advanced Technology, Guangzhou University, Guangzhou, China. He has authored more than 20 publications published by refereed international journals and conferences, e.g., *Information Sciences*, *IEEE TRANSACTIONS ON NETWORK AND SERVICE MANAGEMENT*, *Applied Energy*, and *Energy*. His research interests include network traffic analysis, anomaly detection, and load forecasting.



**Lihua Yin** received the Ph.D. degree from the Harbin Institute of Technology, Harbin, China. She is currently a Professor and an Associate Dean with the Cyberspace Institute of Advanced Technology, Guangzhou University, Guangzhou, China. She has been a Project Leader of the National Natural Science Foundation of China, the National Key research and Development Plan of China, the Major State Basic Research Development Program of China, and the National High-Tech Research and Development Program of China. She has authored

more than 120 papers published by refereed international conferences and journals, e.g., IEEE TRANSACTION ON KNOWLEDGE DATA ENGINEERING, IEEE TRANSACTION ON INDUSTRIAL INFORMATICS, IEEE TRANSACTION ON SOFTWARE ENGINEERING, Computer Security, and Infocom. Her research interests include network security, Internet of Things, secure threat intelligence sharing, and big data. She is an Editorial Board Member of the *Journal of Network and Information Security*.



**Chuang Yin** is currently pursuing the B.Eng. degree in cyberspace security with Guangzhou University, Guangzhou, China. His research focuses on network anomaly detection.



**Jingyi Tan** received the master's degree in computer science from the University of Birmingham, U.K. She is currently an Associate Professor with Guangzhou Xinhua College and serves as the Director of the Information Resource Management Program. Her research focuses on data mining and big data analytics.



**Zhe Sun** received the B.S. degree from the Dalian University of Technology in 2010, the M.S. degree from the University of Science and Technology of China in 2012, and the Ph.D. degree from the University of Chinese Academy of Sciences in 2019. He is an Associate Professor with the Cyberspace Institute of Advanced Technology, Guangzhou University, China. He has authored over 30 publications in international conferences and journals, focusing on privacy-preserving mechanisms for multiparty data sharing and deep learning.

In recognition of his exceptional achievements, he was awarded the prestigious President's Award by the Chinese Academy of Sciences in 2019. Moreover, he serves as a Committee Member of the Big Data Security and Privacy Computing Committee within the Chinese Association for Chinese Information Society.



**Chuhong Ruan** is currently pursuing the B.Eng. degree in cyberspace security with Guangzhou University, Guangzhou, China. His research focuses on malicious traffic detection.



**Xi Luo** received the Ph.D. degree in computer science and technology from the Institute of Information Engineering, Chinese Academy of Sciences in 2019. He is currently an Associate Professor of Network Security with the Cyberspace Institute of Advanced Technology, Guangzhou University, Guangzhou, China. He has authored more than 30 publications published by refereed international journals and conferences, e.g., IEEE TRANSACTION ON INDUSTRIAL INFORMATICS, *Computer Communications*, *Neural Computing and Applications*, *High Performance Computing and Communications*. His research interests include AI-enabled threat analysis, Internet of Things security, and network anomaly detection.

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